Questions

Q1.

Figure 18 shows a person on a skateboard at the top of a ramp.

At P, the person is not moving.

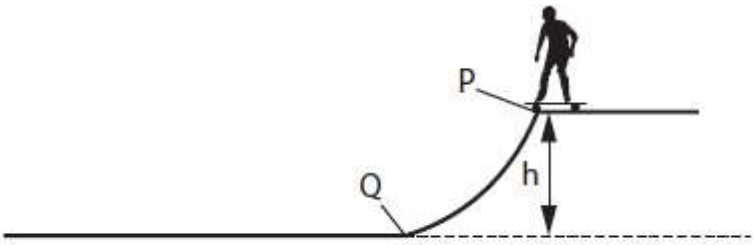


Figure 18

The person would like to start from P again but have a greater velocity at Q.

Suggest **two** ways that this can be achieved.

(2)

1 .....

.....

2 .....

.....

(Total for question = 2 marks)

Q2.

A student lifts a toy car from a bench and places the toy car at the top of a slope as shown in Figure 16.

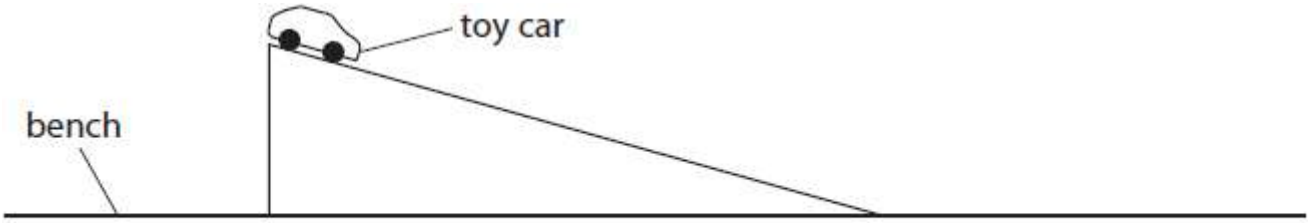


Figure 16

The student needs to develop the experiment to determine the loss in potential energy and the gain in kinetic energy as the toy car is rolling down the slope.

State the other measurements the student must make.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Q3.

A student lifts a toy car from a bench and places the toy car at the top of a slope as shown in Figure 16.

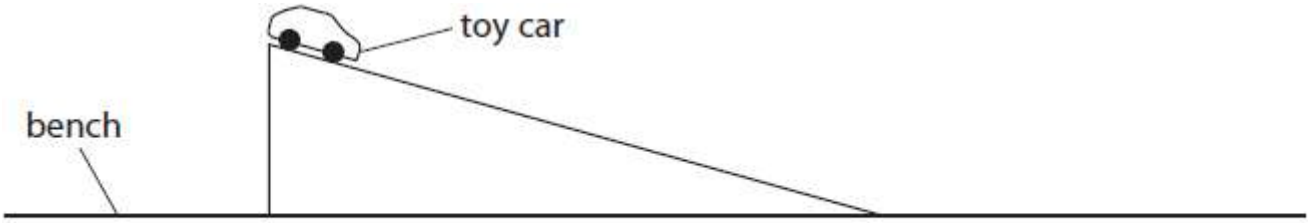


Figure 16

When the toy car rolls down the slope, some energy is transferred to the surroundings as thermal energy.

State how the student could calculate the amount of energy transferred to the surroundings.

(1)

.....

.....

(Total for question = 1 mark)

Q4.

Figure 18 shows a person on a skateboard at the top of a ramp.

At P, the person is not moving.

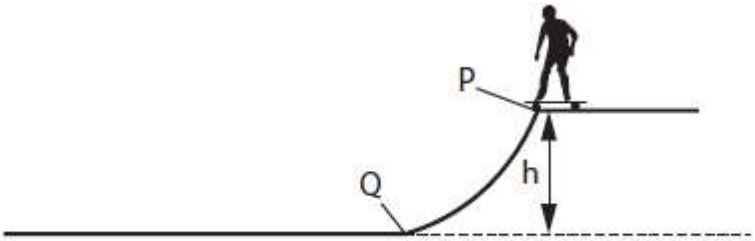


Figure 18

Figure 19 is a diagram that represents energy changes from P to Q.

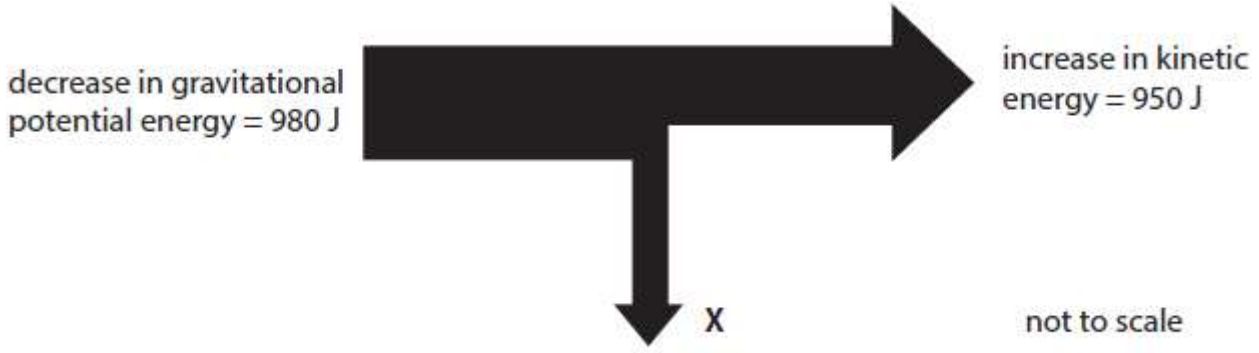


Figure 19

(i) State what is represented by **X**.

(1)

.....

.....

(ii) Calculate the value of **X**.

(1)

value of **X** = ... J

(iii) Calculate the efficiency of the system represented in Figure 19.

(2)

efficiency = ...

(Total for question = 4 marks)

Q5.

Figure 9 shows the renewable energy sources used in the UK in 2015.

Figure 9 is to scale.

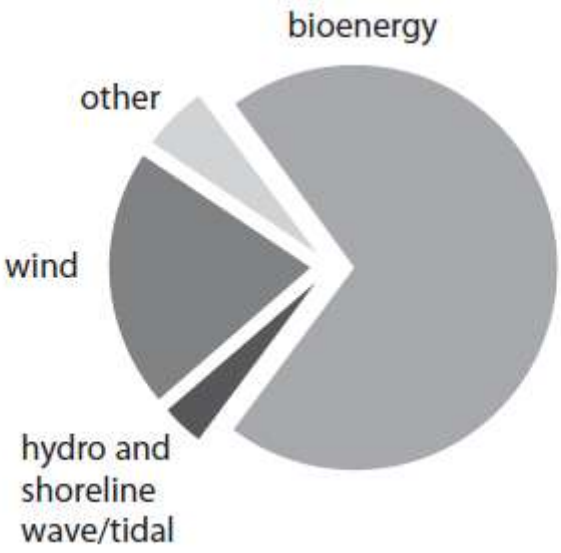


Figure 9

(i) State the energy source that gave the greatest amount of renewable energy for the UK in 2015.

(1)

.....

(ii) Justify your choice of energy source in part (i).

(1)

.....

.....

(iii) State which of these energy sources gave about 20% of the energy from renewable sources for the UK in 2015.

(1)

.....

(Total for question = 3 marks)

Q6.

Figure 10 shows all the energy sources used in Canada in 2014 and a prediction for 2040.

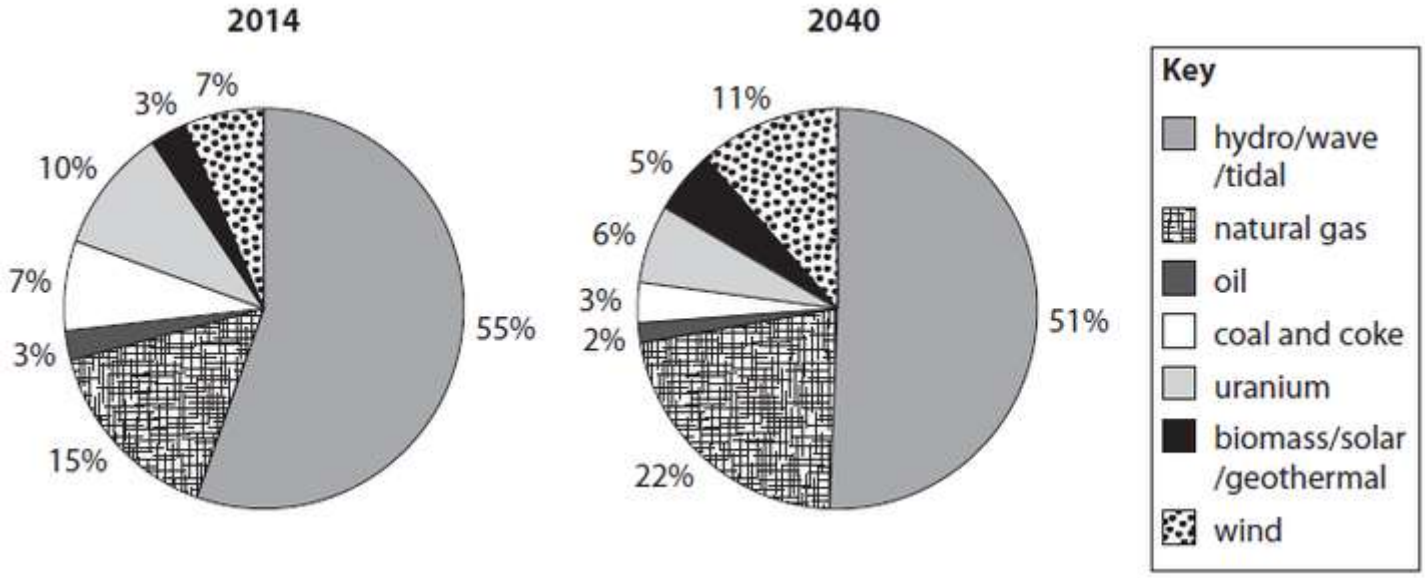


Figure 10

Discuss the effects on the environment of two predicted changes between 2014 and 2040.

(4)

change 1

.....

.....

effect on the environment

.....

.....

change 2

.....

.....

effect on the environment

.....

.....

(Total for question = 4 marks)

Q7.

State **two** non-renewable energy sources.

(2)

1 .....

2 .....

(Total for question = 2 marks)

Q8.

Everyone is exposed to background radiation. Some of this radiation comes from natural sources.
(a) (i) One example of a source of background radiation that does not occur naturally is radiotherapy.
State **one** other source of background radiation that does not occur naturally.

(1)

.....

.....

(ii) Radon gas is a natural source of background radiation.

In some parts of the country, a lot of the background radiation comes from radon gas.

Explain why there is no radon gas in some other parts of the country.

(2)

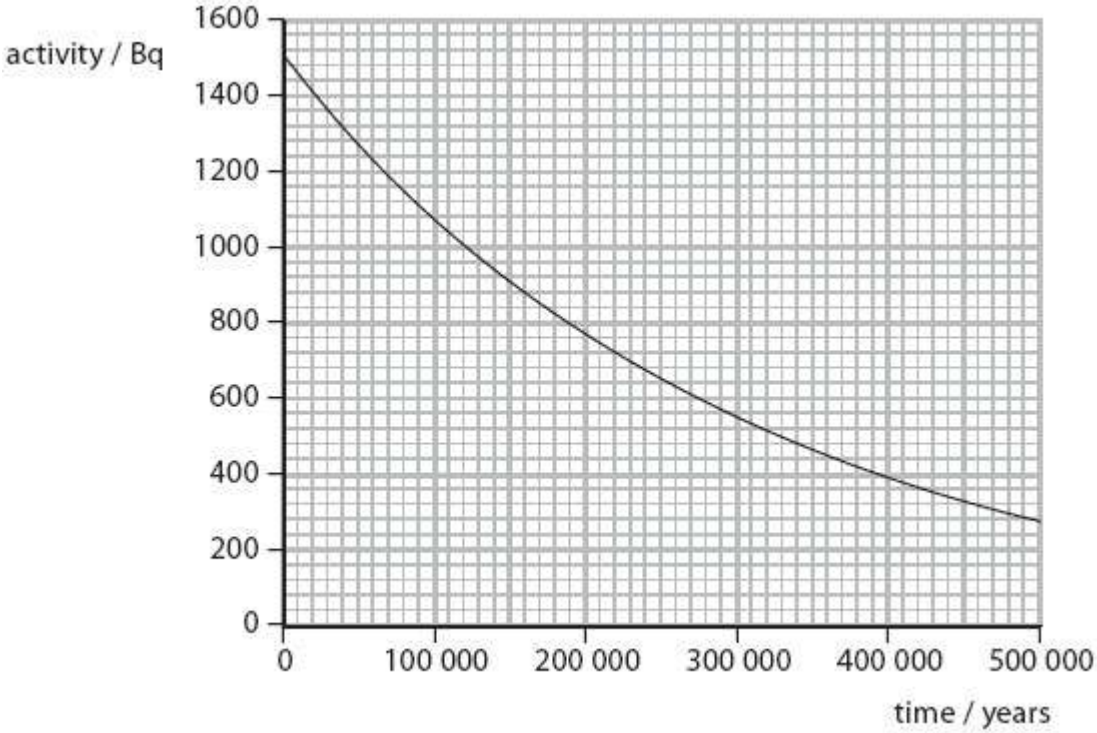
.....

.....

.....

.....

(b) Technetium-99 is one of the radioactive isotopes in nuclear waste.
The graph shows the decay curve for technetium-99.



(i) Use the graph to show that the half-life of technetium-99 is about 200 000 years.

(2)

(ii) Technetium-99 emits beta particles.

Give **one** reason that beta particles can cause harm to people.

(1)

.....

.....

*(c) There are many radioactive isotopes in nuclear waste.
Technetium-99 is just one of these isotopes.

People are worried about how we should deal with nuclear waste.

Explain why it is difficult to deal with nuclear waste safely.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question is 12 marks)

Q9.

Figure 7 shows a skier going down a hill.

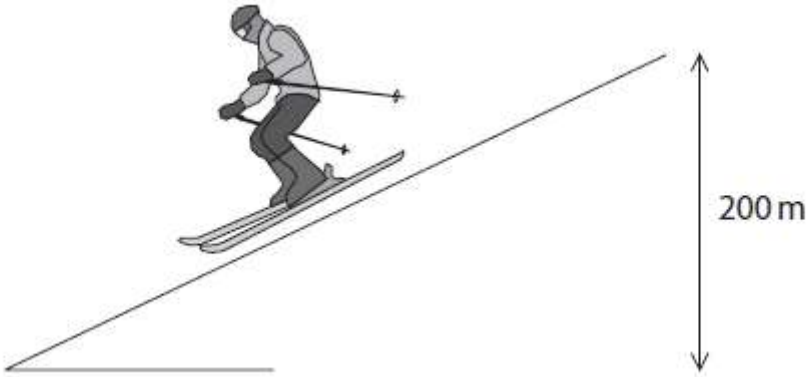


Figure 7

She descends through a vertical height of 200 m.

The skier's mass is 65 kg.

(i) Calculate the change in gravitational potential energy.

Use the equation

$$\Delta GPE = m \times g \times \Delta h$$

Take the gravitational field strength, *g*, as 10 N / kg.

(2)

change in gravitational potential energy = ... J

(ii) At the bottom of the slope her speed was 36 m/s.

Calculate her kinetic energy at the bottom of the slope.
Use the equation

$$KE = \frac{1}{2} \times m \times v^2$$

(3)

kinetic energy = ... J

(Total for question = 5 marks)

Q10.

(i) Which of these would be a typical speed for a racing cyclist travelling down a steep straight slope?

(1)

☐ **A** 0.2 m/s

☐ **B** 2 m/s

☐ **C** 20 m/s

☐ **D** 200 m/s

(ii) A cyclist travels down a slope.
The top of the slope is 20 m vertically above the bottom of the slope.
The cyclist has a mass of 75 kg.

Calculate the change in gravitational potential energy of the cyclist between the top and the bottom of the slope.

The gravitational field strength, g, is 10 N/kg.

(3)

change in gravitational potential energy = ... J

(Total for question = 4 marks)

Q11.

Which one of these is a renewable energy source?

(1)

☐ **A** coal

☐ **B** nuclear

☐ **C** oil

☐ **D** wind

(Total for question = 1 mark)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
	any **two** from reduce air resistance (1) lubricate (wheels) (1) push off (1)	 be more aerodynamic crouch down	**2** **AO1.1**

Q2.

Question number	Answer	Additional guidance	Mark
	(vertical) height of slope (1) mass (of the toy car) (1)	 allow (in this context) weight if no other mark scored allow 1 mark for quoting either equation (Δ)GPE =mgh OR KE = ½ mv^2	**(2)**

Q3.

Question number	Answer	Additional guidance	Mark
	(original) GPE – KE (at bottom) (1)	allow (idea of) input – output allow wrong way round (eg output-input)	**(1)**

Q4.

Question number	Answer	Additional guidance	Mark
(i)	thermal (energy/store) (1)	energy dissipated energy wasted energy lost energy transferred to surroundings energy transferred due to friction energy transferred due to air resistance allow heat (energy) ignore sound/not useful energy	**1** **AO2.1**

Question number	Answer	Additional guidance	Mark
(ii)	30 (J) (1)	accept -30	**1** **AO2.1**

Question number	Answer	Additional guidance	Mark
(iii)	substitution (1) 950 (x100) 980 evaluation (1) 0.97 OR 97%	 allow answers that round to 0.97 or 97% for 2 marks allow 1 mark for answers that round to 97 or 0.97% allow 1 mark for answers of 0.96 or 96% (truncated) award full marks for correct answer without working	**2** **AO2.1**

Q5.

Question Number	Answer	Additional guidance	Mark
(i)	bioenergy	biofuel / biomass	**(1)** AO 3 1b

Question Number	Answer	Additional guidance	Mark
(ii)	largest area / fraction / percentage (idea)	**must be referring to the chart**, not just repeating 4bi stem- can't have greatest/ largest amount by itself	**(1)** AO 3 1b

Question Number	Answer	Additional guidance	Mark
(iii)	wind		(1) AO 3 1b

Q6.

Question Number	Answer	Additional guidance	Mark
	discussion to involve two points each giving change and effect (max 4 marks) some examples: change: biomass-solar-geothermal (fraction) increases (1) effect: e.g. reduces greenhouse gas / CO_2 emissions (1) change: 'wind' (fraction) increases (1) effect: e.g. visual/noise pollution arguments (1) change: 'natural gas' (fraction) increases (1) effect: e.g. contributes to global warming (1) change: 'uranium' (fraction) decreases (1) effect: e.g. less radioactive waste (1)	ignore vague responses such as 'environmentally friendly', less pollution etc. candidates may give positive or negative effects for this change (and for oil) allow decreases (with a correct accompanying effect for 2 marks) accept conserves non-renewables but not just 'more renewable'	**(4)** AO 3 2a AO 3 2b

Q7.

Question Number	Answer	Additional guidance	Mark
	any two sources from: • oil • (natural) gas • coal • nuclear/uranium	accept petrol /diesel for oil accept fossil fuel(s) for any of the first three i.e. fossil fuel and oil or coal or gas scores 1 mark but fossil fuel and nuclear scores 2 marks	(2) AO 1 1

Q8.

	Answer	Acceptable answers	Mark
(a)(i)	any one of X-ray (machines) / smoke alarms/ nuclear/ radioactive waste (1)	nuclear weapons (tests) nuclear power plants (medical) tracers/technetium	**(1)**
(a)(ii)	an explanation linking: comes from granite / rocks (1) none/ less of these (rocks) in some areas (1)	in some areas/Cornwall/Aberdeen the second mark is dependent on the first.	**(2)**
(b)(i)	suitable lines on graph to show halving after about 200 000 years (2) • horizontal line at 750 +or –50 Bq on y-axis to curve (1) • meeting (by eye) vertical line from x-axis between 190,000 years and 230,000 years (1)	use of data from graph to show halving after about 200 000 years 1500/2 =750(Bq) or 1600/2=800(Bq) gives a half-life of 210,000 +or- 20 000 (years)	**(2)**
(b)(ii)	any one of • penetrates/passes through the skin (1) • ionises (1) • damages tissue/ cells/DNA (1) • mutates cells/DNA(1)		**(1)**

	• causes cancer(1)		

Total marks for question = 12

		Indicative Content	Mark
QWC	***(c)**	an explanation which may include some of the following points: properties of nuclear waste radioactivity is dangerous some isotopes in nuclear waste have long half-lives/radioactive for thousands of years products of fission are warm identified radiation from nuclear waste e.g alpha, beta, gamma problems caused by nuclear waste leakage of radioactivity contamination of ground/sea water/lakes /rivers contamination of crops/fish/animals/drinking water harm to humans/cancer/radiation poisoning/ damage to cells/mutation of cells or DNA difficulty in transporting safely/ stolen by terrorists fears of local people solutions for dealing with nuclear waste safely long term storage, underground /under the sea radiation shielding, lead/steel/concrete/ containers, sealed in glass. human safety, radiation suits, using tongs/lead jackets safe location, away from people/remote	**(6)**

		areas/sea cooling, ponds information to persuade local people of safety	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited explanation mentioning at least one point, but without linking, e.g. radioactivity is dangerous ; nuclear waste should be stored underground ; terrorists might steal nuclear waste; • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple explanation mentioning two points with an appropriate linkage e.g. nuclear waste is dangerous and it must be stored underground ; the isotopes in nuclear waste have long half-lives so they must be stored for a long time; • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed explanation mentioning a range of points with appropriate linkages e.g. gamma rays from nuclear waste causes damage to cells so it must be stored away from where people live ; the isotopes in nuclear waste have long half-lives so they must be stored underground or in remote areas; • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	

Q9.

Question Number	Answer	Additional guidance	Mark
(i)	substitution (1) (ΔGPE =) 65 x 10 x 200 evaluation (1) 1.3 x 10^5 / 130 000 (J)	 allow substitution mark with 65000 (g) allow 1 mark for answers that round to 1.3 with any other power of ten do not allow 13000 award full marks for the correct answer without working	(2) AO 2 1

Question Number	Answer	Additional guidance	Mark
(ii)	substitution (1) (KE) ½×65×36(^{2}) squaring (1) 36^2 (=1296) (completing) evaluation (1) 42 120 / 4.2(1) x 10^4 (J)	 using 36 → 1170 (J) OR 36x2 → 2340 (J) scores 2 marks (apply power of ten error as well if occurring e.g. 117000 (J) gets 1 mark) award full marks for the correct answer without working allow 2 marks for answers that round to 4.2 with any other power of ten omitting ½ gives 84240(J) scores 2 marks	(3) AO 2 1

Q10.

Question Number	Answer	Mark
(i)	The only correct answer is **C 20 m/s** A is not correct because 0.2 m/s is too slow B is not correct because 2 m/s is too slow D is not correct because 200 m/s is too fast	(1)

Question Number	Answer	Additional guidance	Mark
(ii)	recall (1) (ΔGPE) = m × g × Δh substitution (1) (ΔGPE =) 75 x 10 x 20 evaluation (1) 15 000 (J)	NO PoT error NO ecf from wrong equation mgh or m x g x h 75 x 10 x 20 scores the first 2 marks accept 14700 (J) from using g = 9.8 (N/kg) award full marks for the correct answer without working	(3)

Q11.

Question number	Answer	Additional guidance	Marks
	D wind		(1)